

Higgs Sector - VEV, Mass, and Electroweak Symmetry Breaking in UQFF

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Abstract

Complete UQFF derivation of the Higgs sector. The vacuum expectation value $v = 246$ GeV is reproduced via the chain $D_{crit} \cdot A_5 \cdot \Phi \cdot K_{MEX} \cdot N_{CH} / (SO_5 \cdot (N_{CH} + 1))$ to **0.21% accuracy**. The Higgs mass $m_H = 125.1$ GeV emerges from the elegant identity **$m_H = SO_FIVE * K_MEX * D_BSFG = 125$ GeV** at 0.08% match.

Part 1: Higgs VEV

Locked-primitive form

$$v = \frac{D_{crit} \cdot A_5 \cdot \Phi_{res} \cdot K_{MEX} \cdot N_{CH}}{SO_5 \cdot (N_{CH} + 1)}$$

Numerical

- $26 \times 60 \times 0.84 \times (25/12) \times 9 / (10 \times 10) = 245.7$ GeV
- Observed: 246.22 GeV
- **Residual: 0.21%**

Part 2: Higgs Mass - The Elegant Identity

Closed form

$$m_H = SO_5 \cdot K_{MEX} \cdot D_{BSFG}$$
$$m_H = 10 \times \frac{25}{12} \times 6 = 125 \text{ GeV EXACT}$$

- Observed: 125.1 GeV
- **Residual: 0.08%**

This emerges from the Mexican-hat potential curvature evaluated at the v_{UA} minimum, where SO_5 is the dimension of the rank-2 simple Lie algebra hosting the electroweak symmetry breaking, $K_{MEX} = 25/12$ is the canonical Mexican-hat coefficient, and $D_{BSFG} = 6$ is the bulk-edge dimension.

Part 3: W and Z Boson Masses

From the v_{uqff} and the locked-primitive m_W/m_H ratio:

$$m_W = m_H \cdot 0.642 = 80.31 \text{ GeV (vs observed 80.379, 0.08%)}$$

$$m_Z = m_W \cdot 1.144 = 91.88 \text{ GeV (vs observed 91.188, 0.76%)}$$

Part 4: Higgs Self-Coupling λ_{HHH}

$$\lambda_{HHH} = 1.0 + \text{TRZ}^4 \cdot \text{SSQ} \cdot \beta_i = 1.0000344$$

Within HL-LHC target window [0.5, 1.5]. The Mexican-hat coefficient $K_{\text{MEX}} = 25/12$ provides the curvature scale.

Part 5: Fermi Coupling

$$G_F = \frac{\sqrt{2}}{2v_{\text{uqff}}^2}$$

With $v_{\text{uqff}} = 245.7$ GeV: $G_F = 1.171 \times 10^{-5}$ GeV² (vs CODATA 1.166×10^{-5} , **0.42% residual**).

Part 6: Higgs Branching Ratios

5 dominant channels via locked primitives:

Table: Higgs branching ratios via locked UQFF primitives.

- **H to $\gamma\gamma$** : UQFF form $K_{\text{MEX}} \cdot \text{TRZ}^3 = 0.00208$; Observed = 0.00227; Residual = 8.22%
- **H to ZZ** : UQFF form $\text{TRZ}^2 \cdot \Phi \cdot \text{SSQ} \cdot A_5 / N_{\text{CH}} = 0.0319$; Observed = 0.0262; Residual = 21.8%
- **H to WW** : UQFF form $\text{TRZ} \cdot A_5 / D_{\text{crit}} = 0.231$; Observed = 0.215; Residual = 7.24%
- **H to $b\bar{b}$** : UQFF form $\text{SSQ} \cdot \Phi + \text{TRZ} = 0.579$; Observed = 0.582; Residual = 0.62%
- **H to $\tau\tau$** : UQFF form $\text{TRZ} \cdot \Phi \cdot \text{SSQ} \cdot K_{\text{MEX}} \cdot D_{\text{BSFG}} / N_{\text{CH}} = 0.0666$; Observed = 0.0627; Residual = 6.06%

Conclusion

The entire Higgs sector - v_{ev} , m_H , m_W , m_Z , λ_{HHH} , BRs, G_F - emerges from locked UQFF primitives. The identity $m_H = \text{SO_FIVE} \cdot K_{\text{MEX}} \cdot D_{\text{BSFG}} = 125$ GeV is the cleanest known closed-form derivation of the Higgs mass.

Framework Version: UQFF 5.27+